

K L Deemed to be University

**CSE-4 -- KLVZA
Course Handout
2026-2027, Odd Sem**

Course Title	: FRONT END DEVELOPMENT FRAMEWORKS FOR WEB APPLICATIONS
Course Code	: 25CS2101
L-T-P-S Structure	: 0-0-2-4
Pre-requisite	: 25SC1203-DATA STRUCTURES
Credits	: 2
Course Coordinator	: Suneetha Bulla
Team of Instructors	: Venkata Padyala, Satya Gouri Arunasri Pabbiseti, Nichenametla Rajesh, CH. M H Saibaba, Nageswara Rao Thota, Ashesh Kinjirapu, Velmurugan A K, Ratna Kanth Nelapati, Suneetha Bulla, Nagendra Rajaboina, Choutapalli Nagaraju, Veeraswamy Ammisetty, Phani Chintamaneni, Dineshnath Gopinath, Nyakapu Rajender, Lakshmi Kodavali, Pallabothula Ramesh, Pallabothula Ramesh, nuthakki praveena, SATYA KEERTHI CH, SETTI RAJESWARI, Seetharam R, Seetharam R, annamaraju thanuja, Thella Priyanka , BALAKRISHNA BANGARU, Lakshmi Ramani Burra, K. VaniSri, K. VaniSri, RAMAKRISHNA REGULAGADDA, Shaik Sanjeera, RAVISANKAR MALLADI, KOLLU NAIDU, ANUSHA GUDISE, BANDLA NIROSHA, PANGULURI PADMAVATHI, YELISELA RAJESH, VENKATA SRIRAM BHEEMISETTY, sailakshmi tirumalasetty, Tayar Yerramsetty, Tayar Yerramsetty, NIMMAGADDA MURALIKRISHNA, Neerukattu Rao, Neerukattu Rao, CHERUKUPALLI KUMAR, CHERUKUPALLI KUMAR, Spandana Mande, JALLI BABU, Seelam Rao, Vijay R, ANITHA PALAKAYALA, Pavani Chitrapu
Teaching Associates	:
Syllabus :	CO-1: Introduction to Git and Version Control, Git Installation and Configuration, Git Life Cycle Repository Creation, Remote Repository basic Operations, Branching and Merging, Conflict Resolution, Git History Management, .gitignore, and Git Workflow. Introduction to HTML5 structure, Semantic HTML5 tags, Forms and input types, Media elements (audio, video, images), Advanced HTML5 elements (section, aside, nav, header, footer), HTML5 form enhancements (placeholder, required, autofocus, pattern) CO-2 CSS Syntax, Types of CSS, Selectors, Box Model, Positioning Techniques, Flexbox and Grid Layouts, Responsive Web Design Principles, Media Types and Queries, CSS Transitions, Transformations, and Keyframe Animations. Introduction to JavaScript, Syntax, Data Types, Operators, Functions, Control Structures, Objects, Arrays, Sets, Maps, DOM Manipulation, Event Handling, ES6+ Features, Promises, Async/Await, Fetch API, Form Validation, Error Handling, Local and Session Storage, Modular JavaScript, Classes and Objects, JSON Processing, and Debugging Tools. CO-3 Introduction to React, JSX, Functional Components, Props and State, Event Handling, Conditional and List Rendering, React Hooks (useState, useEffect), Forms, Component Communication, React Router, Application Structure, Reusable Components, Atomic Design, Redux Toolkit, Material UI, Tailwind CSS, and Deployment with Vite. CO-4 Advanced React Concepts (Context API, Custom Hooks, Lazy Loading, Error Boundaries), React Testing with Jest and React Testing Library, Introduction to Node.js, npm and HTTP Server Development, Express.js Setup and Routing, RESTful API Development (GET/POST), Middleware, Security Practices, Input Validation, and Data Sanitization. CO-5 Introduction to Next.js, React vs

	Next.js, File-Based and Dynamic Routing, API Routes, SSG and SSR, Data Fetching, Image Optimization, SEO, Middleware and NextAuth.js Authentication, MongoDB and Mongoose Integration, Environment Variables, Tailwind CSS, Deployment on Vercel, and Full-Stack Capstone Project Development. CO - 6 Analyze production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented frontend workflow.
Text Books :	1 Learning React — Modern Patterns for Developing React Apps, Alex Banks & Eve Porcello O'Reilly 2020 (2nd Ed.) 2 React Design Patterns and Best Practices Carlos Santana Roldán Packt 2023 (3rd Ed.) 3 Effective TypeScript — 62 Specific Ways to Improve Your TypeScript Dan Vanderkam O'Reilly 2024 (2nd Ed.) 4 Programming TypeScript Boris Cherny O'Reilly 2019
Reference Books :	1. Inclusive Components Heydon Pickering, Smashing Magazine, 2019 2. JavaScript — The Definitive Guide, David Flanagan O'Reilly, 2020 (7th Ed.) 3. Refactoring UI Adam Wathan & Steve Schoger, Self-published, 2018
Web Links :	1. Git and GitHub for Beginners - https://www.coursera.org/learn/introduction-git-github 2. JavaScript: The Advanced Concepts - https://www.udemy.com/course/advanced-javascript-concepts/ 3. Next.js & React - The Complete Guide - https://www.udemy.com/course/nextjs-react-the-complete-guide/
MOOCS :	1. Meta Front-End Developer Professional Certificate - https://www.coursera.org/professional-certificates/meta-front-end-developer 2. Developing Front-End Apps with React - https://www.coursera.org/learn/developing-frontend-apps-with-react 3. Front-End Development Libraries V8 - https://www.freecodecamp.org/learn/front-end-development-libraries/ 4. React - The Complete Guide (incl. Next.js, Redux) - https://www.udemy.com/course/react-the-complete-guide-incl-redux/?couponCode=MT260714G1
Course Rationale :	Modern front-end development requires strong engineering and architectural thinking beyond framework syntax. This course adopts an architecture-first approach to explain why front-end frameworks exist and how they address complexity, performance, and scalability challenges. It emphasizes declarative UI design, component-based architectures, state management, and asynchronous data handling. Learners gain the ability to reason about rendering, performance, and maintainability in production-grade front-end systems.
Course Objectives :	1. To equip students with skills in version control and front-end development using Git, HTML5, CSS, and JavaScript. 2. To develop proficiency in building interactive web applications using React and modern JavaScript features 3. To enable students to create scalable back-end services and full-stack applications using Node.js, Express.js, and Next.js. 4. To foster problem-solving and debugging skills through hands-on projects and real-world web development practices.
Global Certifications :	GitHub Copilot Certification (Exam GH-300) - https://learn.github.com/certification/COPILOT
Industry Specific Tools :	1. VS Code - https://code.visualstudio.com/download 2. NodeJS - https://nodejs.org/en/download
Sector Skill Council :	National Skill Development Corporation – The Front-End Developer job role under IT-ITeS SSC, with QP Code SSC/Q1401, is defined as NSQF Level 6.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Analyze Git-based version control workflows, repository management, and HTML5 semantic structures to develop standards-compliant and collaborative web applications.	PO1,PO2,PSO1	2
CO2	Develop dynamic web applications using JavaScript, ES6+ features, and DOM manipulation.	PO3,PSO1,PO1	3
CO3	Build interactive user interfaces using React, including state management and routing.	PO3,PSO1,PO1	3
CO4	Implement back-end services using Node.js, Express.js and integrate testing and security practices.	PSO1,PO1,PO3	3
CO5	Apply data-fetching with TanStack Query, forms with React Hook Form + Zod, routing with React Router — to build production-ready single-page applications.	PO3,PO5,PSO1,PO1	3
CO6	Analyze production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented frontend workflow.	PO3,PO5,PSO1,PO1,PO6	4

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
CO1	2	Btl-1 Remember modern front-end architectures, UIengineering constraints,rendering pipelines, andcomponent-drivenengineering principles thatform the foundation ofReact and allcontemporaryframeworks	Btl-2 Understand modern front-end architectures, UIengineering constraints,rendering pipelines, andcomponent-drivenengineering principles thatform the foundation ofReact and allcontemporaryframeworks.		
CO2	3	Btl-1 Remember JavaScript(ES6+) and TypeScriptengineering patterns—including modular design,immutability, functionalcomposition, asyncpipelines—to constructscalable front-endarchitectures.	Btl-2 Understand JavaScript(ES6+) and TypeScriptengineering patterns—including modular design,immutability, functionalcomposition, asyncpipelines—to constructscalable front-endarchitectures.	Btl-3 Understand JavaScript(ES6+) and TypeScriptengineering patterns—including modular design,immutability, functionalcomposition, asyncpipelines—to constructscalable front-endarchitectures.	
CO3	3	Btl-1 Remember Reactapplications	Btl-2 Understand Reactapplications	Btl-3 Build React applicationsusing	

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
		usingcomponent-drivendevelopment, state andprops architecture, side-effect management,reconciliationfundamentals, andcontrolled UI patterns.	usingcomponent-drivendevelopment, state andprops architecture, side-effect management,reconciliationfundamentals, andcontrolled UI patterns.	component-drivendevelopment, state andprops architecture, side-effect management,reconciliationfundamentals, andcontrolled UI patterns.	
CO4	3	Btl-1 Remember front-endapplications using statemanagement strategies,asynchronous data flows,API integration layers,and architectural patternslike lifting state,composition, andcontainer–presenterdesign.	Btl-2 Understand front-endapplications using statemanagement strategies,asynchronous data flows,API integration layers,and architectural patternslike lifting state,composition, andcontainer–presenterdesign.	Btl-3 Apply front-endapplications using statemanagement strategies,asynchronous data flows,API integration layers,and architectural patternslike lifting state,composition, andcontainer–presenterdesign.	
CO5	3	Btl-1 Remember routingstrategies, formengineering, accessibility(a11y), performancetuning, and renderingoptimizations to createhighly usable andperformant Reactapplications.	Btl-2 Understand routingstrategies, formengineering, accessibility(a11y), performancetuning, and renderingoptimizations to createhighly usable andperformant Reactapplications.	Btl-3 Apply routing strategies,form engineering,accessibility (a11y),performance tuning, andrendering optimizations tocreate highly usable andperformant Reactapplications	
CO6	4	Btl-1 Remember production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented frontend workflow.	Btl-2 Understand production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented frontend workflow.	Btl-3 Apply production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented frontend workflow.	Btl-4 Analyze production-React practices — Vite build pipeline, Vercel deployment, accessibility (WCAG AA), performance (memo, code splitting), the AI-augmented

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
					frontend workflow.

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO1	Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
PO3	Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations
PO5	Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PSO1	An ability to design and develop software projects as well as Analyze and test user requirements.

Lecture Course DELIVERY Plan:

NO Delivery Plan Exists

Lecture Session wise Teaching – Learning Plan

No Session Plans Exists

Tutorial Course DELIVERY Plan:

NO Delivery Plan Exists

Tutorial Session wise Teaching – Learning Plan

No Session Plans Exists

Practical Course DELIVERY Plan:

Tutorial Session no	Topics	CO-Mapping
1	Create a Git Repository and Develop Features Using Multiple Branches, Merge Feature Branches into the Main Branch	CO1
2	Deploy Static Web Application into git Pages using git workslows	CO1
3	Design the homepage of an e-commerce website using CSS Flexbox, Grid, and animation techniques.	CO1
4	Develop a responsive Student Admission Form with CSS Grid layout that works seamlessly across desktop, tablet, and mobile devices.	CO2
5	Develop a web-based Student Result Management System using html,css and JavaScript.	CO2
6	Develop a Weather Information Dashboard using modern JavaScript features. Fetch weather data from an API using Fetch API	CO2
7	Develop a React application with Login, Dashboard, and Profile modules.	CO3
8	Building a Course Management Dashboard Using React, Redux Toolkit, and Material UI	CO3
9	Designing a University Event Management Portal Using Atomic Design, Material UI, and Tailwind CSS	CO3
10	Create React based application authentication with a Node.js Express backend.	CO4
11	Developing a Secure Student Portal Using Context API and Express Middleware	CO4
12	Optimizing and Testing an Employee Management Portal	CO4

Tutorial Session no	Topics	CO-Mapping
13	build a static React + Vite app with 3 components (Header, Card, Footer); deploy to Vercel; observe build vs dev mode.	CO5
14	build a small to-do app with custom useTodos hook; persist to localStorage via custom useLocalStorage hook; test with React Testing Library.	CO5
15	convert the M2 to-do app to TypeScript; type all props, state, custom hooks; resolve all "any" warnings.	CO6
16	build a small filter-sidebar UI in three ways — Context, Zustand, Redux Toolkit; reflect on the boilerplate-vs-clarity trade-off	CO6

Practical Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Create a Git Repository and Develop Features Using Multiple Branches, Merge Feature Branches into the Main Branch

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Think / Pair / Share
40	Implementing the Concepts	3	LTC	Quiz/Test Questions

SESSION NUMBER : 2

Session Outcome: 1 Deploy Static Web Application into git Pages using git workslows

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 3

Session Outcome: 1 Design the homepage of an e-commerce website using CSS Flexbox, Grid, and animation techniques.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Think-Pair-Share
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just-in-time Teaching

SESSION NUMBER : 4

Session Outcome: 1 Develop a responsive Student Admission Form with CSS Grid layout that works seamlessly across desktop, tablet, and mobile devices.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	Talk	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 5

Session Outcome: 1 Develop a web-based Student Result Management System using html,css and JavaScript.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 6

Session Outcome: 1 Develop a Weather Information Dashboard using modern JavaScript features. Fetch weather data from an API using Fetch API

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 7

Session Outcome: 1 Develop a React application with Login, Dashboard, and Profile modules.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 8

Session Outcome: 1 Building a Course Management Dashboard Using React, Redux Toolkit, and Material UI

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 9

Session Outcome: 1 Designing a University Event Management Portal Using Atomic Design, Material UI, and Tailwind CSS

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 10

Session Outcome: 1 Create React based application authentication with a Node.js Express backend.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	Talk	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 11

Session Outcome: 1 Developing a Secure Student Portal Using Context API and Express Middleware

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	Talk	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 12

Session Outcome: 1 Optimizing and Testing an Employee Management Portal

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 13

Session Outcome: 1 build a static React + Vite app with 3 components (Header, Card, Footer); deploy to Vercel; observe build vs dev mode.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Discussing the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 14

Session Outcome: 1 build a small to-do app with custom useTodos hook; persist to localStorage via custom useLocalStorage hook; test with React Testing Library.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	2	Talk	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 15

Session Outcome: 1 convert the M2 to-do app to TypeScript; type all props, state, custom hooks; resolve all "any" warnings.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 16

Session Outcome: 1 build a small filter-sidebar UI in three ways — Context, Zustand, Redux Toolkit; reflect on the boilerplate-vs-clarity trade-off

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

Skilling Course DELIVERY Plan:

Skilling session no	Topics/Experiments	CO-Mapping
1	Install and configure Git, create a local repository, and integrate it with a remote repository for version control and collaboration.	CO1
2	Perform remote repository operations, create and manage branches, merge changes, and resolve merge conflicts in a collaborative Git environment.	CO1
3	Manage Git history, configure .gitignore files, and implement an efficient Git workflow for project development and version control.	CO1
4	Design the Home Page of an E-Commerce Website Using HTML5 Structure and Semantic Elements	CO1
5	Develop Product Catalog and Multimedia Advertisement Pages Using HTML5 Media Elements	CO1
6	Design Customer Registration and Checkout Forms Using Advanced HTML5 Form Features	CO1
7	Converting an HTML Table-Based Application Form into a CSS Grid Layout	CO2
8	Developing an E-Commerce Website Header and Navigation Menu Using Flexbox	CO2
9	Designing E-Commerce Website Body Content and Footer with Grid, Animations, and Responsive Design	CO2
10	Implementing User Registration, Login, and Form Validation for an E-Commerce Website	CO2

Skilling session no	Topics/Experiments	CO-Mapping
11	Developing Dynamic Product Catalog and Shopping Cart Functionality	CO2
12	Integrating Product Search, API-Based Product Loading, and Order Tracking Dashboard	CO2
13	Converting an HTML/CSS Grid-Based Web Page into React Functional Components	CO3
14	Implementing Login and Signup Forms Using Props and State	CO3
15	Building Dynamic Product Listing Using React Hooks	CO3
16	Implementing Navigation and Page Redirection Using React Router	CO3
17	Creating Reusable UI Components Using Atomic Design, Material UI, and Tailwind CSS	CO3
18	Implementing Global State Management using redux	CO3
19	Migrating Login and Signup Authentication from Local Storage to Node.js Express Server	CO4
20	Creating User Dashboard and Profile APIs Using Express.js	CO4
21	Implementing Context API and Custom Hooks for Authentication Management	CO4
22	Implementing Lazy Loading and Error Boundaries for Dashboard Modules	CO4
23	Securing Registration and Login APIs Using Middleware, Validation, and Sanitization	CO4
24	Testing the Complete React and Node.js Authentication System	CO4
25	TanStack Query (Rule 39 contemporary blend) — useQuery for reads, useMutation for writes, query invalidation, optimistic updates.	CO5

Skilling session no	Topics/Experiments	CO-Mapping
26	Why TanStack Query replaced useEffect-for-fetching — caching, background refetching, status flags, request deduplication.	CO5
27	Error and loading states — the canonical UI patterns; error boundaries.	CO5
28	The Vite build pipeline — dev server with HMR, production build with tree-shaking and code-splitting.	CO6
29	Deployment to Vercel — connect GitHub repo, every push deploys, preview URLs for PRs.	CO6
30	Accessibility in React (WCAG AA) — focus management on route changes, ARIA on custom components, semantic HTML inside JSX.	CO6

Skilling Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Install and configure Git, create a local repository, and integrate it with a remote repository for version control and collaboration.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Attendance and Recap	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 2

Session Outcome: 1 Perform remote repository operations, create and manage branches, merge changes, and resolve merge conflicts in a collaborative Git environment.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 3

Session Outcome: 1 Manage Git history, configure .gitignore files, and implement an efficient Git workflow for project development and version control.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 4

Session Outcome: 1 Design the Home Page of an E-Commerce Website Using HTML5 Structure and Semantic Elements

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 5

Session Outcome: 1 Develop Product Catalog and Multimedia Advertisement Pages Using HTML5 Media Elements

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 6

Session Outcome: 1 Design Customer Registration and Checkout Forms Using Advanced HTML5 Form Features

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 7

Session Outcome: 1 Converting an HTML Table-Based Application Form into a CSS Grid Layout

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 8

Session Outcome: 1 Developing an E-Commerce Website Header and Navigation Menu Using Flexbox

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just-in-time Teaching
40	Discussing the Topic	3	LTC	Just-in-time Teaching
50	Implementing the Concepts	3	LTC	Just-in-time Teaching

SESSION NUMBER : 9

Session Outcome: 1 Designing E-Commerce Website Body Content and Footer with Grid, Animations, and Responsive Design

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 10

Session Outcome: 1 Implementing User Registration, Login, and Form Validation for an E-Commerce Website

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just-in-time Teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 11

Session Outcome: 1 Developing Dynamic Product Catalog and Shopping Cart Functionality

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 12

Session Outcome: 1 Integrating Product Search, API-Based Product Loading, and Order Tracking Dashboard

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 13

Session Outcome: 1 Converting an HTML/CSS Grid-Based Web Page into React Functional Components

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 14

Session Outcome: 1 Implementing Login and Signup Forms Using Props and State

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 15

Session Outcome: 1 Building Dynamic Product Listing Using React Hooks

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	3	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 16

Session Outcome: 1 Implementing Navigation and Page Redirection Using React Router

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 17

Session Outcome: 1 Creating Reusable UI Components Using Atomic Design, Material UI, and Tailwind CSS

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	3	LTC	Just in-time teaching
40	Discussing the Topic	4	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 18

Session Outcome: 1 Implementing Global State Management using redux

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Introduction	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 19

Session Outcome: 1 Migrating Login and Signup Authentication from Local Storage to Node.js Express Server

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 20

Session Outcome: 1 Creating User Dashboard and Profile APIs Using Express.js

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Demonstrating the Concept	3	LTC	Just in-time teaching

SESSION NUMBER : 21

Session Outcome: 1 Implementing Context API and Custom Hooks for Authentication Management

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 22

Session Outcome: 1 Implementing Lazy Loading and Error Boundaries for Dashboard Modules

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 23

Session Outcome: 1 Securing Registration and Login APIs Using Middleware, Validation, and Sanitization

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	LTC	Just in-time teaching
40	Explaining the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 24

Session Outcome: 1 Testing the Complete React and Node.js Authentication System

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	3	LTC	Just in-time teaching
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 25

Session Outcome: 1 TanStack Query (Rule 39 contemporary blend) — useQuery for reads, useMutation for writes, query invalidation, optimistic updates.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	Talk	Just in-time teaching
40	Explaining the Topic	3	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 26

Session Outcome: 1 Why TanStack Query replaced useEffect-for-fetching — caching, background refetching, status flags, request deduplication.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	2	Talk	Just-in-time Teaching
40	Explaining the Topic	3	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 27

Session Outcome: 1 Error and loading states — the canonical UI patterns; error boundaries.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 28

Session Outcome: 1 The Vite build pipeline — dev server with HMR, production build with tree-shaking and code-splitting.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 29

Session Outcome: 1 Deployment to Vercel — connect GitHub repo, every push deploys, preview URLs for PRs.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
40	Discussing the Topic	3	LTC	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

SESSION NUMBER : 30

Session Outcome: 1 Accessibility in React (WCAG AA) — focus management on route changes, ARIA on custom components, semantic HTML inside JSX.

Time(min)	Topic	BTL	Teaching- Learning Methods	Active Learning Methods
10	Recap and Attendance	1	Talk	Just in-time teaching
40	Explaining the Topic	2	PPT	Just in-time teaching
50	Implementing the Concepts	3	LTC	Just in-time teaching

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDEDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
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COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	-- -	-- -	---	---	--	--	-- -	---	---
	Tutorial	-- -	-- -	---	---	--	--	-- -	---	---
	Lab	-- -	-- -	---	---	--	--	-- -	---	---
	Skilling	-- -	-- -	---	---	V-S11,V-S11,V-S12,V-S12,V-	V-S11,V-S11,V-S12,V-S12,V-	-- -	---	---

	Hour	1	2	3	4	5	6	7	8	9
						S13,V-S13,V-S14,V-S14,V-S15,V-S15	S13,V-S13,V-S14,V-S14,V-S15,V-S15			
Tue	Theory	-- -	-- -	--	--	---	---	-- -	---	---
	Tutorial	-- -	-- -	--	--	---	---	-- -	---	---
	Lab	-- -	-- -	--	--	---	---	-- -	---	---
	Skilling	-- -	-- -	V-S11,V-S11,V-S12,V-S12,V-S13,V-S13,V-S14,V-S14,V-S15,V-S15	V-S11,V-S11,V-S12,V-S12,V-S13,V-S13,V-S14,V-S14,V-S15,V-S15	---	---	-- -	---	---
Wed	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Thu	Theory	-- -	-- -	---	---	---	---	-- -	--	--
	Tutorial	-- -	-- -	---	---	---	---	-- -	--	--
	Lab	-- -	-- -	---	---	---	---	-- -	V-S11,V-S11,V-S11,V-S12,V-S12,V-S12,V-S13,V-S13,V-S13,V-S14,V-S14,V-S14,V-S15,V-S15,V-S15	V-S11,V-S11,V-S11,V-S12,V-S12,V-S12,V-S13,V-S13,V-S13,V-S14,V-S14,V-S14,V-S15,V-S15,V-S15
	Skilling	-- -	-- -	---	---	---	---	-- -	--	--
Fri	Theory	-- -	-- -	--	--	---	---	-- -	---	---
	Tutorial	-- -	-- -	--	--	---	---	-- -	---	---

	Hour	1	2	3	4	5	6	7	8	9
	Lab	-- -	-- -	--	--	---	---	-- -	---	---
	Skilling	-- -	-- -	V-S61,V-S61,V-S62,V-S62,V-S63,V-S63,V-S64,V-S64,V-S65,V-S65	V-S61,V-S61,V-S62,V-S62,V-S63,V-S63,V-S64,V-S64,V-S65,V-S65	---	---	-- -	---	---
Sat	Theory	-- -	-- -	--	--	---	---	-- -	---	---
	Tutorial	-- -	-- -	--	--	---	---	-- -	---	---
	Lab	-- -	-- -	--	--	---	---	-- -	---	---
	Skilling	-- -	-- -	V-S61,V-S61,V-S62,V-S62,V-S63,V-S63,V-S64,V-S64,V-S65,V-S65	V-S61,V-S61,V-S62,V-S62,V-S63,V-S63,V-S64,V-S64,V-S65,V-S65	---	---	-- -	---	---
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplementary course handouts, including special lectures and discussions, will be planned and the schedule will be notified accordingly.

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Minutes)	CO1	CO2	CO3	CO4	CO5	CO6
End Semester Summative Evaluation Total= 40 %	Hackathon-Final Review	Weightage	22		90	0.5	0.5	0.5	6.5	7	7
		Max Marks	100			16	16	16	16	16	20
	Lab End Semester Exam	Weightage	18		90	0.5	0.5	0.5	5.5	5.5	5.5
		Max Marks	50			8	8	8	8	8	10
In Semester Formative Evaluation Total= 24 %	Continuous Evaluation -Project	Weightage	12		90	2	2	2	2	2	2
		Max Marks	100			16	16	16	16	16	20
	Continuous Evaluation - Lab Exercise	Weightage	12		90	2	2	2	2	2	2
		Max Marks	100			16	16	16	16	16	20
In Semester Summative Evaluation Total= 36 %	Global Certification Completion	Weightage	20		90	5	5	5	5		
		Max Marks	100			25	25	25	25		
	Lab In sem-II	Weightage	8		90	2	2	2	2		
		Max Marks	50			12.5	12.5	12.5	12.5		
	Lab In sem-I	Weightage	8		90	2	2	2	2		
		Max Marks	50			12.5	12.5	12.5	12.5		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course

In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Plagiarism is treated as a serious academic offense, and the university enforces a strict zero-tolerance policy to uphold academic integrity. For a first offense, minor cases may result in a warning and reduced marks, while major cases may lead to a zero grade and mandatory academic integrity training. A second offense will result in a zero grade, formal reprimand, and academic probation, whereas a third offense may lead to suspension and a permanent record of misconduct. Repeated violations can result in expulsion, and in group work, all members are held accountable unless responsibility is clearly established.

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Venkata Padyala	S	61-B	-	-	-	-
Satya Gouri Arunasri Pabbiseti	P	12-B	-	-	-	-
Nichenametla Rajesh	P	61-B	-	-	-	-
CH. M H Saibaba	P	61-C	-	-	-	-
Nageswara Rao Thota	P	13-C	-	-	-	-
Ashesh Kinjirapu	P	63-B	-	-	-	-
Velmurugan A K	S	62-B	-	-	-	-
Ratna Kanth Nelapati	S	65-B	-	-	-	-
Suneetha Bulla	P	63-C	-	-	-	-
Nagendra Rajaboina	S	11-B	-	-	-	-
Choutapalli Nagaraju	P	15-B	-	-	-	-
Veeraswamy Ammisetty	S	14-B	-	-	-	-
Phani Chintamaneni	P	64-B	-	-	-	-
Dineshnath Gopinath	S	62-B	-	-	-	-
Nyakapu Rajender	P	64-C	-	-	-	-
Lakshmi Kodavali	P	62-B	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Pallabothula Ramesh	P	61-A,11-A	-	-	-	-
Pallabothula Ramesh	S	11-A,61-A	-	-	-	-
nuthakki praveena	P	65-B	-	-	-	-
SATYA KEERTHI CH	S	63-B	-	-	-	-
SETTI RAJESWARI	S	64-B	-	-	-	-
Seetharam R	P	62-A,12-A	-	-	-	-
Seetharam R	S	62-A,12-A	-	-	-	-
annamaraju thanuja	S	13-B	-	-	-	-
Thella Priyanka	P	14-B	-	-	-	-
BALAKRISHNA BANGARU	P	62-C	-	-	-	-
Lakshmi Ramani Burra	P	15-C	-	-	-	-
K. VaniSri	P	63-A,13-A	-	-	-	-
K. VaniSri	S	63-A,13-A	-	-	-	-
RAMAKRISHNA REGULAGADDA	P	11-B	-	-	-	-
Shaik Sanjeera	S	63-B	-	-	-	-
RAVISANKAR MALLADI	S	15-B	-	-	-	-
KOLLU NAIDU	S	61-B	-	-	-	-
ANUSHA GUDISE	P	65-C	-	-	-	-
BANDLA NIROSHA	S	12-B	-	-	-	-
PANGULURI PADMAVATHI	S	64-B	-	-	-	-
YELISELA RAJESH	S	65-B	-	-	-	-
VENKATA SRIRAM BHEEMISSETTY	P	14-C	-	-	-	-
sailakshmi tirumalasetty	S	15-B	-	-	-	-
Tayar Yerramsetty	P	14-A,64-A	-	-	-	-

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Tayar Yerramsetty	S	14-A,64-A	-	-	-	-
NIMMAGADDA MURALIKRISHNA	S	14-B	-	-	-	-
Neerukattu Rao	P	15-A	-	-	-	-
Neerukattu Rao	S	15-A	-	-	-	-
CHERUKUPALLI KUMAR	P	65-A	-	-	-	-
CHERUKUPALLI KUMAR	S	65-A	-	-	-	-
Spandana Mande	P	13-B	-	-	-	-
JALLI BABU	S	12-B	-	-	-	-
Seelam Rao	S	11-B	-	-	-	-
Vijay R	P	12-C	-	-	-	-
ANITHA PALAKAYALA	P	11-C	-	-	-	-
Pavani Chitrapu	S	13-B	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes by reviewing the prescribed video lectures in advance and carrying the required textbook(s) or study material(s) as specified by the Course Faculty.

NOTICES

All course-related notices and communications will be shared through the official institutional email.

Signature of COURSE COORDINATOR

(Suneetha Bulla)(6655)(EL&GE)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of CSE-1

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal)

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